

Jaswanthi B

AI Engineer

713-364-4583 | jaswanthi@itjobinbox.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | CA, USA

SUMMARY:

AI Engineer with 7 years of experience designing scalable AI, machine learning, cloud data engineering, and analytics solutions across insurance, energy, research, and enterprise domains. Experienced in building RAG-based LLM applications, multi-agent AI systems, cloud-native data pipelines, and production-grade MLOps/LLMOps workflows using AWS, Databricks, Python, and SQL. Proven track record of improving operational efficiency through intelligent automation, semantic search, and advanced analytics while ensuring security, compliance, and high system reliability. Strong collaborator with expertise in translating complex business requirements into scalable, data-driven solutions that deliver measurable business impact.

SKILLS:

Programming Languages: Python, SQL, PySpark, Bash

AI / Machine Learning & Orchestration: LangChain, Amazon Bedrock, Claude 3.5 Sonnet, NeMo Guardrails, Multi-Agent Systems, TruLens, MLflow, NLP, Vector Embeddings

Vector Search & Databases: Milvus, Databricks Unity Catalog, Amazon Redshift, Microsoft SQL Server, PostgreSQL

Cloud Platforms & Infrastructure: AWS (SageMaker, Glue, Lambda, S3, Comprehend, CloudWatch, CloudFormation), Databricks, Terraform, Docker, CI/CD (Jenkins, Git)

Data Engineering & Big Data: Apache Spark, Apache Kafka, Apache Airflow, SSIS, Data Pipelines, ETL/ELT Design

Data Quality & Analytics: Tableau, Power BI, Advanced SQL Optimization, Hybrid Search (BM25 + Semantic)

EDUCATION:

Master of Science (M.S.) in Data Science

University of Houston, USA | Dec 2024

Bachelor of Technology (B. Tech) in Computer Science and Engineering

Institute of Aeronautical Engineering, India | Jun 2022

EXPERIENCE:

Allstate, IL | Jun 2025 – Present | AI Engineer

- Engineered a RAG-based LLM pipeline on AWS SageMaker and LangChain to automatically synthesize multi-document claim histories into structured adjuster summaries, reducing average claim review times by 32% across auto and property insurance lines.
- Implemented dynamic claim communications drafting using Amazon Bedrock and Claude 3.5 Sonnet, saving 22,000+ adjusters an average of 40 minutes daily on written correspondence while maintaining high contextual accuracy.
- Designed real-time PII redaction and guardrails using AWS Comprehend and custom NeMo Guardrails, preventing sensitive financial/medical data leakage and ensuring 100% compliance with state insurance privacy regulations.
- Optimized vector index retrieval on Databricks Unity Catalog and Milvus using hybrid BM25-semantic search, improving document retrieval precision by 27% and reducing LLM hallucination rates on complex policy clauses.
- Established an automated LLMOps evaluation workflow with MLflow and TruLens, enabling continuous benchmarking of model latency and accuracy across 50,000+ daily generated correspondence drafts in production.

University of Houston, Houston, TX | Jan 2023 – May 2025 | Research Assistant – Data Science & AI

- Developed a multi-agent scholarly research intelligence system using LLM-powered AI agents to automate author disambiguation, citation analysis, co-authorship mapping, and academic network analytics across large-scale publication datasets.
- Built AI agents for author identity resolution and metadata enrichment using NLP pipelines, embedding models, and external research APIs, significantly reducing manual validation effort in scholarly data processing workflows.
- Designed and deployed a RAG-based conversational AI chatbot enabling researchers to query publication metrics, collaboration networks, gender disparity analysis, and research trends through natural language interactions.
- Built backend data pipelines and vector search workflows for semantic document retrieval, embedding generation, and contextual knowledge retrieval using Python-based AI orchestration frameworks.
- Tutored 150+ students in Python, SQL, and web technologies while maintaining Tableau dashboards and automated analytics pipelines for academic performance and attendance monitoring.

Capgemini, Bengaluru, India | Mar 2022 – Dec 2022 | Cloud Data Engineer

Client - Chevron Corporation

- Engineered scalable data ingestion pipelines using AWS Glue, Apache Spark, Amazon S3, and Apache Kafka to consolidate drilling, production, and equipment telemetry from multiple enterprise systems, reducing daily data availability time by 38% for engineering and operations teams.
- Developed robust ETL workflows with PySpark and AWS Glue to cleanse, transform, and load structured and semi-structured operational data into Amazon Redshift, improving data quality by 32% while decreasing analytical query response times by 35%.
- Implemented automated data quality validation, workflow orchestration, and monitoring using Apache Airflow, AWS Lambda, and CloudWatch, reducing pipeline failures by 28% and improving reliability across business-critical production analytics workloads.
- Automated infrastructure provisioning and CI/CD deployments using Terraform, CloudFormation, Jenkins, and Git while collaborating with cloud, security, and business teams to deliver secure, compliant data platform enhancements, reducing deployment effort by 40% and accelerating release cycles.

Virtual Infotech Solution, India | Oct 2019 – Mar 2022 | Data Analyst

- Designed a centralized reporting solution by integrating SQL Server databases, Excel reports, and CRM exports using SQL, Python, and Power BI, reducing manual reporting efforts by 70% while providing executives with a single source of trusted business insights.
- Developed optimized ETL workflows using SQL Server Integration Services (SSIS) and Python to cleanse, validate, and transform transactional data from multiple business units, improving data accuracy by 32% and significantly reducing inconsistencies across enterprise reports.
- Built interactive Power BI dashboards featuring sales trends, customer acquisition, revenue performance, inventory movement, and KPI scorecards, enabling department heads to identify performance gaps 40% faster and support weekly operational planning meetings.
- Optimized complex SQL queries, stored procedures, and indexed reporting tables to improve large dataset processing efficiency, reducing dashboard refresh time by 45% while supporting growing business data volumes without impacting report reliability.
- Collaborated with business analysts, finance teams, sales managers, and client stakeholders to translate reporting requirements into analytical solutions, increasing dashboard adoption by 35% through intuitive visualizations and standardized business metrics.
- Automated recurring report generation and data validation processes using Python scripts, SQL jobs, and scheduled refresh pipelines, reducing reporting errors by 30% while ensuring timely delivery of daily and monthly business performance reports.
- Documented reporting standards, data definitions, and validation procedures while supporting Agile sprint activities, user acceptance testing, and production deployments, improving knowledge transfer and reducing post-release reporting issues by 25% across multiple client engagements.

CERTIFICATIONS:

AWS Certified Solutions Architect – Associate (SAA-C03) [\[Link\]](#)

Infosys Certified Software Programmer [\[Link\]](#)